intranet servers or as small database servers—cost no more than Rs 75,000.

To test the best, we received six servers from five vendors—two reputed for their servers, namely HP and Acer (who sent in two for this test), and the other three from Zenith, Wipro and PCS respectively. All of them were based on an Intel platform; most of them featured a Pentium 4 processor, the exception being the Acer G520, which had an Intel Xeon. The price varied from Rs 40,000 to over Rs 1 lakh.

It is not possible to pin-point a perfect server, since its performance varies dramatically with the applications you run on it. However, it is possible is to test overall performance, from which a buyer can guage a certain machine's performance under the expected workload.

The Heart Of The Beast: The Chipset

For any computer, there are four vital subsystems, namely the CPU, memory, I/O and graphics. Apart from the graphics, the others form a very important foundation for any server under consideration.

The chipset dictates the server's features, and the strength and efficiency of the CPU-memory subsystem. Know your chipset and you can make a wise investment. Intel's i875P and E7210 are commonly used by vendors for their servers.

The i875P is referred to as a Workstation chipset by Intel, whereas the E7210 is called a server chipset. Both are nearly identical in terms of features, and both extend support to the new Pentium 4 processor over an 800 MHz FSB (front-side bus), with speeds touching 3.2 GHz. They also support dual-channel DDR memory modules; the bandwidth this affords is rather critical from a server point of view.

Both the chipsets support Gigabit Ethernet via a dedicated bus called the Communication Streaming Architecture (CSA). This should help alleviate PCI bottlenecks, since in other chipsets, the Ethernet ports are tied to the PCI bus, which has a low bandwidth.

The most notable difference between the two chipsets is within the I/O sub-system. The E7210 natively supports PCI-X slots—64-bit avatars of the regular 32-bit PCI; a PCI-X bus is generally used by SCSI controllers, which demand higher bandwidth. The i875P supports PAT—Performance Acceleration Technology (*See box 'Jargon Buster'*).

Acer Altos G310; based on a i875P chipset

The chipset generally dictates the features of a server and the strength and efficiency of the CPU-memory subsystem. Know your chipset and you can make a wise investment

The AGP 8X bus, which is not supported by the E7210, is the reason for calling the i875P a Workstation chipset.

Also, since the E7210 is designed to be a server chipset, reliability would need to be an important aspect of the chipset itself. An i875P-based server is a cheaper option than one based on the E7210.

A third chipset in this comparison was the E7320 from Intel. This brand-new server chipset from Intel is similar to the E7210 save for two major differences. First, this chipset supports Intel Xeon processors. Moreover, motherboards based on this chipset can support two Xeon processors.

Secondly, the inclusion of high-bandwidth PCI-Express buses gives it the edge. This makes the peripheral sub-system fast, robust and future-proof. Solutions for this bus, however, are not yet mature, nor are they widely available.

The i875P chipset is a good choice for medium loads, offers good features, has a decent performance and is well-priced. The E7210 offers a stable server platform with a robust I/O sub-system at a reasonable price.

The E7320 chipset offers the best features and, since it's designed as a server chipset, should serve you well.

The Contenders

Two servers—Acer's Altos G310 and PCS Siirus Compact V—use Intel's i875p chipset. Zenith's Data Server, the HP ProLiant ML-110 and Wipro's NetPower 1120 use the E7210 as their backbone, whereas Acer's Altos G520 had the feature-packed E7320 chipset.

The Processors That Power Them

The Intel Pentium4 2.8 GHz processor, based on the Prescott core—with 1 MB of L2 cache running over an 800 MHz FSB—formed the processing powerhouse of servers using the i875P chipset.

Which Server Is Best For You?

Before deciding on a server for your enterprise, there are a few questions that you must have the answers to:

What is the role of the server?
Is the server a multimedia application server, streaming audio or video over a network? Is it a less demanding Web Proxy or an I/O transfer-dependent fileserver? Based on the server's designated role, you will need the appropriate memory and processing power.

Does it have room to grow?
Your business needs are likely to outgrow your server offerings. Before buying a server, ensure that the setup you're looking at has enough room for growth. Look for sufficient number of extra drive bays, memory slots, PCI slots, etc.

What about security?
In the event of a crash or a virus infection, how does the server stack up in the recovery department? Does it have an efficient RAID setup? A RAID setup can mirror the same set of data across hard disks to minimise the risk of data loss. Equally important are signed memory modules with ECC (Error Checking and Control) support.

How good is the support and warranty offered?
When a server goes down, how quickly can the dealer's support staff help you bring it back up? Does the warranty cover all crucial components?

How much can you spend?
Lower cost doesn't translate to cost-effectiveness. If the server has enough processing power, but not enough memory, the bottleneck created will amount to wastage of CPU power. So pick a well balanced, well-rounded configuration that fits in your budget.